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Effect of 0.01% atropine eye drops combined with different optical treatments to control low myopia in Chinese children

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ABSTRACT

Purpose: To investigate the efficacy and safety of 0.01% atropine (AT) in combination with different optical treatments for controlling myopia in Chinese children.

Methods: This retrospective study analyzed 341 Chinese children aged 6–11 years with myopia between −0.50 D and −3.0 D between January 2022 and May 2023. The fast-progressing, myopic children received three optical treatments combined with 0.01 % atropine: 75 children with single-vision spectacles and atropine (SV + AT), 162 children with defocus-incorporated multi-segment spectacles and atropine (DIMS + AT), or 104 children with orthokeratology and atropine (OK + AT). The changes in spherical equivalent refraction (SER), axial length (AL), intraocular pressure (IOP), and amplitude of accommodation (AMP) were observed at 6-month and 1-year intervals.

Results: After controlling for baseline variables, at 6 months, the increase in adjusted AL was significantly greater in the SV + AT group than in the DIMS + AT group (difference = 0.13 mm, 95 % CI: 0.07–0.20, $P < 0.05$) and in the OK + AT group (difference = 0.09 mm, 95 % CI: 0.09–0.17, $P < 0.05$). A more significant progression in adjusted SER was also observed in the SV + AT group than in the DIMS group (difference = −0.20D, 95 % CI: −0.29 to −0.11, $P < 0.05$). At 12 months, the greatest increase in adjusted AL was observed in the SV + AT group, with a statistically significant difference of 0.24 mm (95 % CI: 0.19–0.29, $P < 0.05$) compared with the DIMS group and a difference of 0.19 mm (95 % CI: 0.13–0.25, $P < 0.05$) compared with the OK + ST group. Similarly, a more significant progression in adjusted SER was observed in the SV + AT group than in the DIMS group (difference = −0.36 D, 95 % CI: −0.48 to −0.24, $P < 0.05$).

Conclusions: This study suggested that 0.01% atropine combined with DIMS or orthokeratology may be viable for controlling low myopia in fast-progressing, myopic children.

1. Introduction

Myopia in children and adolescents is a global public health problem. The prevalence of myopia has been increasing worldwide, mainly due to urbanization and the education process. It is estimated that nearly half of the global population will be affected by myopia by the year 2050 [1]. The rise in myopia is of concern because of its associated health risks and economic burdens [2,3]. High myopia significantly increases the risk of vision-threatening complications such as retinopathy, glaucoma, macular hemorrhage, and retinal choroidal atrophy [4–8]. Given the potential long-term consequences of myopia and its impact on public health, there is growing recognition of the need for myopia control strategies.

Several methods and interventions have been studied for myopia

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control, and some common approaches include orthokeratology (Ortho-K) lenses, multifocal contact lenses, defocus-incorporated multi-segment spectacles (DIMS), and atropine (AT) [9–12]. Compared with single-vision spectacles (SVs), the common feature of these myopia optical interventions is that they provide central correction for distance vision and induce peripheral myopic retinal defocus simultaneously [9,13]. Ortho-K lenses are specially designed gas-permeable contact lenses that are worn overnight, and they have been shown to slow the progression of myopia in some children [14–16]. It has been shown that DIMS lenses have a 9-mm central monovision zone surrounded by a honeycomb zone consisting of multiple + 3.5D myopic defocus segments [9]. Compared with single-vision lenses, DIMS lenses can delay the progression of myopia by 52 % in two years [9].

Atropine has been widely used to control myopia [17]. ATOM2 has been shown to be effective and safe when different concentrations of atropine are compared, indicating that 0.01 % atropine is the optimal concentration for clinical use [18,19]. In addition, the ATLAS study reported that long-term use of topical atropine for childhood myopia control is safe [20]. Studies have shown that the combination of low-dose atropine and Ortho-K showed add-in effects in controlling myopia [21–25]. Recently, some studies have reported the efficacy of DIMS in combination with 0.01 % atropine for myopia control [21,22,26,27]. The current study evaluated the efficacy and safety of 0.01 % atropine in combination with DIMS, Ortho-K, or SV to slow myopia progression in fast-progressing, myopic Chinese children and to provide clinical evidence for the development of individualized myopic treatments in the future.

2. Materials and methods

2.1. Patient enrollment

In this retrospective study, children with low myopia who attended outpatient clinics at Shenyang Sinqi Eye Hospital between January 2022 and May 2023 were analyzed. Fig. 1 illustrates the flow diagram of the study design. A total of 658 myopic children between the ages of 6 and 11 years who had previously undergone optical treatments—such as single-vision spectacles, DIMS, or Ortho-K—were screened in this study. If the children experienced axial length growth above 0.2 mm every six months or 0.40 mm per year (matching changes with SER), they were classified as "fast progressors" and were subsequently prescribed previous optical treatments combined with 0.01 % atropine.

Specifically, the clinician first considered the follow-up examination at 6 months; if the axial length growth was more than 0.2 mm every 6 months, a combination with 0.01 % atropine was initiated. If not, the clinician proceeded to the 12-month follow-up. If axial length growth exceeded 0.40 mm per year, a combination with 0.01 % atropine was initiated. Otherwise, the subsequent six-month follow-up commenced. During follow-up visits, the spectacles were adjusted as needed to ensure optimal visual acuity and biovision. Parents or legal guardians of the children were fully informed about the goal and methodology of the current study and gave their written informed consent for the preservation and use of the data for scientific reasons. The Shenyang Sinqi Eye Hospital's Institutional Research Ethics Committee (2024–010-01) approved this study, which complied with the Declaration of Helsinki's principles.

2.2. Inclusion and exclusion criteria

Children receiving 0.01 % atropine eye drops in combination with different optical treatments and meeting the following criteria were included in this retrospective study: (1) aged 6–11 years; (2) refractive error measured with an autorefractor before and after cycloplegia (0. 5 % compound tropicamide), and the participants were free of ocular and other systemic diseases affecting visual acuity and myopia progression according to a series of eye examinations; (3) −0.50D to −3.00D SER, astigmatism≥-1.50DC, anisometropia less than −1.00DS; (4) 0.01 % atropine eye drops were used once a day before they slept. The 0.01 % atropine eye drops should be applied at least 15 min before wearing the Ortho-K, and no other eye drops should be used within 15 min; (5) best corrected visual acuity (BCVA, logMAR) ≤ 0 in both eyes; (6) intraocular pressure (IOP) 10 to 21 mmHg in both eyes; and (7) follow-up every three months during treatment and complete relevant examinations.

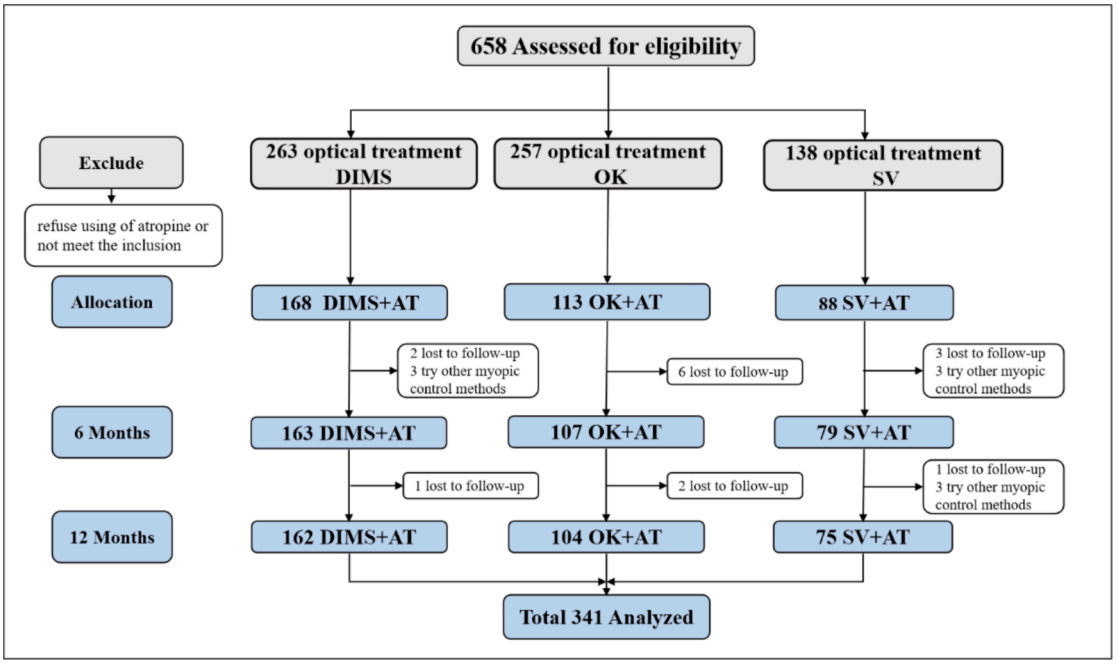

Fig. 1. A flow diagram of the study design. DIMS + AT, 0.01 % atropine drops combined with Defocus Incorporated Multiple Segments spectacle lens; OK + AT, 0.01 % atropine drops combined with orthokeratology lens; SV + AT, 0.01 % atropine drops combined with single vision spectacle lens.

2.3. Eye-related examinations

Refraction was performed via an autorefractor (AKR-1, NIDEK, Japan). Optimal visual acuity was examined with a logMAR visual acuity chart (VSK-VC-Y, Weishikang, Guangzhou, Guangdong, China). Axial length was analyzed with an IOL Master optical biometer (IOL-Master700, Carl Zeiss, Germany). The IOP was examined with a noncontact tonometer (NT-530P, NIDEK, Japan). The amplitude of accommodation (AMP) was measured via the push-up method, in which an examiner holds a visual marker 40 cm from the subject's eyes under corrected vision. The subject focuses on the marker, and the examiner moves the marker closer to it at 5 cm/sec until the image becomes blurred. The distance at which this occurs is recorded, and the reciprocal is the AMP for both eyes. Corneal staining was visualized via fluorescence staining under a slit lamp with a cobalt blue filter.

2.4. Orthokeratology design

A total of 104 children received a combination of Ortho-K and atropine. Of those, 72 (69.23 %) wore VST-designed Ortho-K, and 32 (30.76 %) wore CRT-designed Ortho-K with Jessen factors of + 0.75D. The clinician did not modify the diameter of the BOZD; thus, a diameter of 6.0 mm was typically applied. The positioning of the Ortho-K was centered, and the topography map showed that the defocusing ring was centered without noticeable deviation. There were no severe adverse reactions during the wearing period.

2.5. Statistical analysis

To avoid potential correlation problems between the two eyes, the analysis focused on data from the right eye only. The difference in AL and SER from baseline at 6 months and 1 year was used as the outcome variable in the evaluation of the validity index. The Shapiro–Wilk test was used to test the normality of the data. Comparisons were made via ANOVA (when the variance was homogeneous) or the rank sum test (when the variance was unequal), and comparisons between groups were made via Dunn's method when there were differences between groups. The analysis was performed on complete case data without imputation for missing data and dropouts. To control for baseline age, sex, SE, and AL, a linear mixed model analysis was used. Analyses were performed using SPSS 23.0 and R V.4.0.3 (R Foundation for Statistical Computing). The results were considered statistically significant if the two-sided P value was ≤0.05.

3. Results

3.1. Subject profile

A total of 658 myopic children were retrospectively selected from the outpatient clinic of Shenyang Sinqi Eye Hospital between January 2022 and May 2023. Three groups consisting of a total of 341 children were identified: 162 children in the DIMS + AT group, 104 in the OK + AT group, and 75 in the SV + AT group. To eliminate any potential correlation issues between the two eyes, this analysis focused exclusively on data from the right eye.

As shown in Table 1, the mean (SD) ages were 8.48 (1.45), 9.64 (1.02), and 8.63 (1.00) years for the DIMS + AT, OK + AT, and SV + AT groups, respectively. The initial SERs were −1.63 (0.71) D, −2.16 (0.70) D, and −1.14 (0.84) D, respectively. The initial ALs were 24.08 (0.71), 24.42 (0.81), and 23.79 (0.54) mm, respectively. Table 2 presents the SER and AL changes observed at 6 and 12 months for the three groups.

3.2. Changes in SER

Table 2 presents the adjusted mean changes in SER and AL among the three groups, analyzed with linear mixed models, which were consistent with the data before adjustment (Supplementary Table 1). After controlling for baseline variables, compared with that at baseline, the mean adjusted change in SER at 6 months was −0.09 D (95 % CI, −0.14 to −0.04) in the DIMS + AT group and −0.29 D (95 % CI, −0.36 to −0.22) in the SV + AT group. At the 12-month follow-up, the mean adjusted change in SER was −0.22 D (95 % CI, −0.28 to −0.15) in the DIMS + AT group and −0.58 D (95 % CI, −0.68 to −0.48) in the SV + AT group (Fig. 2). Fig. 3 shows the percentage of myopia progression in both groups. In the DIMS + AT group, 87 % of the children had myopia progression greater than −0.5 D, and 13 % had progression less than −0.75 D, while in the SV + AT group, 49.3 % had myopia progression greater than −0.5 D and 50.7 % less than −0.75 D.

Table 1
Baseline demographic data of all the subjects.

clinical value	DIMS + AT(n = 162)	OK + AT(n = 104)	SV + AT(n = 75)	F/2	P
Age (SD)	8.48 (1.45)	9.64 (1.02)	8.63 (1.00)	29.8	<0.001
Female, NO. (%)	79(48.8 %)	52(50 %)	45(60 %)	2.75	0.25
Male, NO. (%)	83(51.2 %)	52(59 %)	30(40 %)		
AL (mm)	24.08 (0.71)	24.42 (0.81)	23.79 (0.54)	17.62	<0.001
SER (D)	−1.63(0.71)	−2.16(0.70)	−1.14 (0.84)	42.66	<0.001

DIMS + AT, 0.01 % atropine combined with defocus-incorporated multi-segment spectacles.
OK + AT, 0.01 % atropine combined with orthokeratology.
SV + AT, 0.01 % atropine combined with single-vision spectacles.
SER, spherical equivalent refraction.
AL, axial length.

3.3. Changes in AL

After controlling for baseline variables, at the 6-month follow-up, AL increased from baseline in all three groups. The adjusted mean change in AL was 0.09 mm (95 % CI, 0.05–0.12) in the DIMS + AT group, 0.13 mm (95 % CI, 0.08–0.18) in the OK + AT group, and 0.22 mm (95 % CI, 0.16–0.27) in the SV + AT group (Table 2). At 12 months of follow-up, the adjusted mean AL increase in the SV + AT group was 0.40 (95 % CI, 0.36–0.44) mm, which was greater than that in the DIMS + AT group was 0.16 mm (95 % CI, 0.14–0.19), whereas the OK + AT group presented an adjusted mean AL increase of 0.21 mm (95 % CI, 0.17–0.24). The adjusted change in AL was reduced in the DIMS + AT and OK + AT groups compared with the SV + AT group. The adjusted increase was significantly greater in the SV + AT group than in the DIMS + AT group (difference = 0.13 mm, 95 % CI: 0.07 to 0.20, $P < 0.05$) and in the OK + AT group (difference = 0.09 mm, 95 % CI: 0.01 to 0.17, $P < 0.05$). (Table 2, Fig. 4). However, there was no significant difference between the DIMS + AT and OK + AT groups at the 6- or 12-month follow-ups.

Fig. 5 shows the percentage increase in AL for the three groups over a 12-month follow-up. The SV + AT group presented the highest prevalence of children with AL growth of more than 0.3 mm (66.20 %), followed by the OK + AT group (22.10 %), and the DIMS + AT group presented the lowest prevalence (11.70 %).

3.4. Safety assessment

In the safety evaluation, no statistically significant differences in IOP were found between the three groups at baseline and follow-up or in the within-group pairwise comparisons between the two follow-up visits and baseline (Table 3). Similarly, no differences in AMP were found among the three groups either at baseline or at the first follow-up. All three groups demonstrated a reduction in AMP, and the changes observed at follow-up were not significantly different from those observed within the groups (Table 4). With respect to corneal staining, 2

Table 2
Adjusted changes in the SER and AL (from baseline) at 6- and 12-month intervals in the three groups.

	DIMS + AT(n = 162)	OK + AT(n = 104)	SV + AT(n = 75)	Mean difference (SV-DIMS)	Mean difference (SV-OK)	Mean difference (OK-DIMS)
Time/visit	Changes in AL(mm)					
6-month	0.09 (0.05, 0.12)	0.13 (0.08, 0.18)	0.22 (0.16, 0.27)	0.13 (0.07, 0.20) *	0.09 (0.01, 0.17) *	−0.04 (−0.1, 0.02)
12-month	0.16 (0.14,0.19)	0.21 (0.17, 0.24)	0.40 (0.36, 0.44)	0.24 (0.19, 0.29) *	0.19 (0.13, 0.25) *	−0.04 (−0.09, 0.01)
Time/visit	Changes in in SER (D)	0.23				
6-month	−0.09 (0.14, −0.04)	NA	−0.29 (−0.36, −0.22)	−0.20 (−0.29, −0.11)*	NA	NA
12-month	−0.22 (−0.28, −0.15)	NA	−0.58 (−0.68, 0.48)	−0.36(−0.48, −0.24)*	NA	NA

DIMS + AT, 0.01 % atropine combined with defocus-incorporated multi-segment spectacle.
OK + AT, 0.01 % atropine combined with orthokeratology.
SV + AT, 0.01 % atropine combined with single-vision spectacles.
SER, spherical equivalent refraction.
AL, axial length.
* P<0.05.

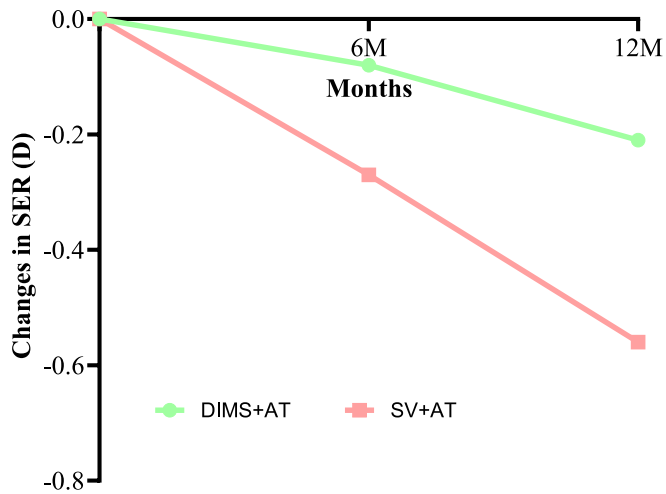

Fig. 2. Adjusted changes in adjusted spherical equivalent refraction (SER) from baseline to 6- and 12-month intervals in the DIMS and SV groups.

(1.19 %), 2 (2.66 %), and 13 (12.50 %) of the three groups presented grade I corneal staining. Grade II corneal staining was observed in only three eyes in the OK + AT group (Table 5).

4. Discussion

Currently, optical interventions combined with low-concentration atropine are being utilized to control the progression of adolescent myopia. This study included fast-progressing, myopic children (with AL growth of more than 0.40 mm per year) with optical treatments alone. After 1 year of follow-up, both the DIMS + AT and OK + AT groups were significantly more effective at controlling AL growth in children than the SV + AT group, and the differences between the DIMS + AT and OK + AT groups were also not significant. In recent years, promising results have been reported from the combination of Ortho-K and 0.01 % atropine in controlling myopia [28–30]. However, further clinical evaluation is needed to confirm the efficacy and safety of the combination of DIMS and 0.01 % atropine [21,26]. This study aimed to contribute to this evaluation by comparing the efficacy of different optical treatments combined with 0.01 % atropine.

Notably, this study paid attention to the efficacy of myopia control in children aged 6–11 years. The younger the individual is, the greater the risk of high myopia in the future and the greater the difficulty of myopia control. In 2019, a longitudinal study in Singapore revealed that SER progressed faster and lasted longer in early-onset myopia (defined as ≤ 8 years) than in late-onset myopia (defined as > 10 years) [31]. Studies

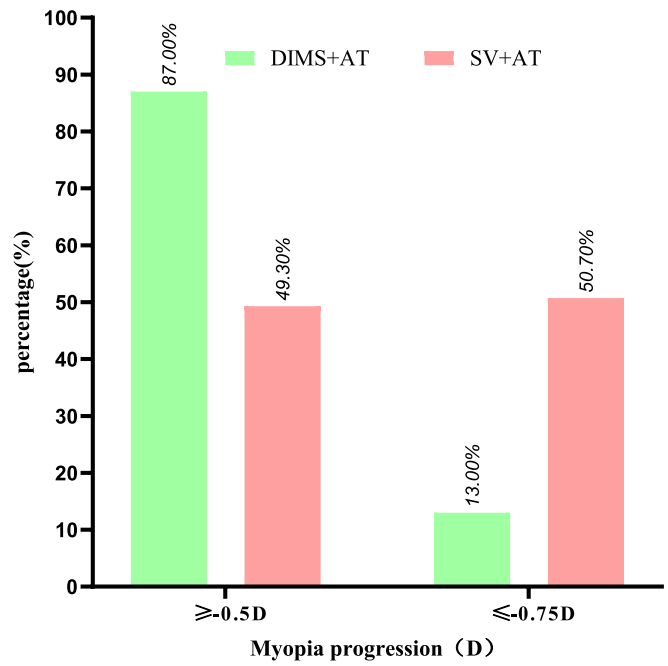

Fig. 3. Percentages of myopia progression in the DIMS + AT and SV + AT groups. DIMS + AT, 0.01 % atropine drops combined with a defocus-incorporated multiple-segment spectacle lens; SV + AT, 0.01 % atropine drops combined with a single-vision spectacle lens.

have shown that the risk of high myopia is relatively high in children with myopia onset during early school years [32–35]. Additionally, a study [36] reported that younger children may require higher concentrations of atropine eye drops (0.05 %) to achieve the same reduction in myopic progression as older children receive lower concentrations (0.025 % or 0.01 %). Similarly, in studies of combined Ortho-K therapy [23,25,37–39], younger age is an essential factor that affects its efficacy.

Research has shown that a change in AL of 0.20 mm/year can be used as a predictor of myopia progression in nonmyopic children aged 7–10 years [40]. In this study, the decision to combine previous treatments with 0.01 % atropine was based on AL progression of 0.20 mm/6 months or 0.40 mm/year when using optical treatments only. Several studies have shown that the severity of growth depends on age, and children less than 11 years of age commonly exhibit fast, progressive elongation in axial length, which appears to stabilize after the age of 12 [31,41,42] As shown in Fig. 5, the proportion of children with AL growth less than 0.3 mm was 88.30 % in the DIMS + AT group, 77.90 % in the OK + AT group, and 33.80 % in the SV + AT group. The current study revealed

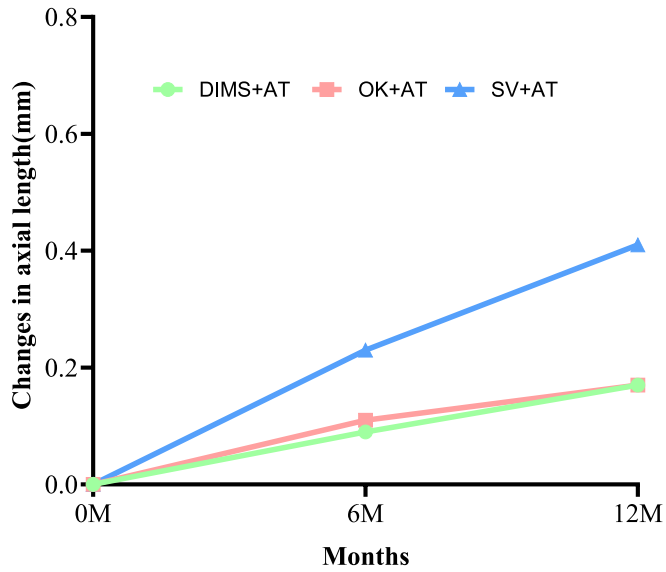

Fig. 4. Adjusted changes in adjusted axial length (AL) from baseline to 6- and 12-month intervals in the DIMS, SV, and OK groups.

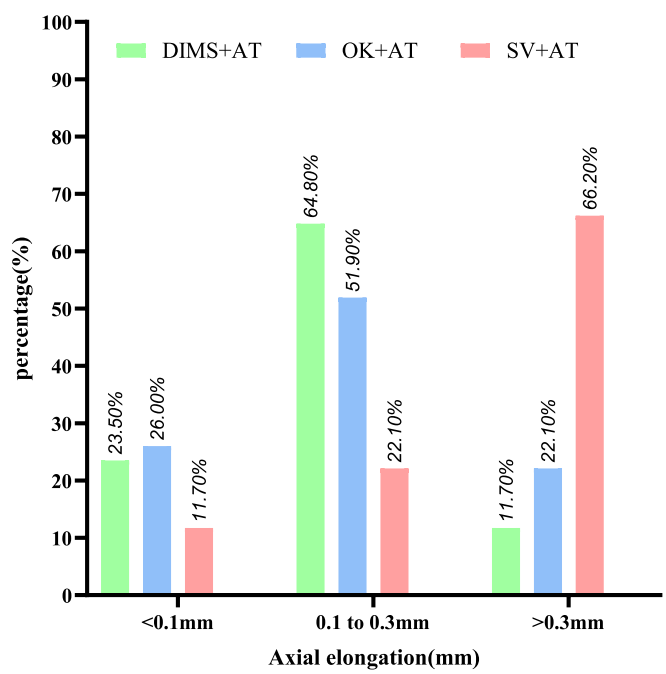

Fig. 5. Percentage increase in AL over 12 months. DIMS + AT, 0.01 % atropine drops combined with Defocus Incorporated Multiple Segments spectacle lens; OK + AT, 0.01 % atropine drops combined with orthokeratology lens; SV + AT, 0.01 % atropine drops combined with single vision spectacle lens.

that the combination of 0.01 % atropine with a DIMS lens or Ortho-K can significantly control myopia progression. Interestingly, the proportions of children whose AL growth was less than 0.1 mm and greater than 0.3 mm were greater in the OK + AT group than in the DIMS + AT group. Additionally, the SV + AT group had the greatest number of children (66.20 %) with AL growth of more than 0.3 mm in 1 year, which suggests that 0.01 % atropine alone without retinal-image defocus may not be sufficient to achieve an adequate therapeutic effect, which is consistent with the findings of others [43,44].

To date, one of the most critical issues for controlling AL growth is retinal image defocus [45–48]. Studies have shown that the amount of defocus produced by Ortho-K on the retina is determined by mRCRP (relative corneal refractive power) [49,50]. With a peak mRCRP less than −3.0 D, a subject has an approximately 80 % chance to achieve adequate control [51], whereas the control effect of a refraction greater than −3.0 D is significantly reduced, which may partly explain the poor efficacy of some myopic children with a smaller baseline SER when wearing the Ortho-K. However, the DIMS lens has multiple annular multifocal zones of relatively positive power (+3.50 D) to control myopia independent of the baseline SER [9], which may contribute to myopia control in the DIMS + AT group with low-grade myopia.

Furthermore, for myopia control with poor response, especially in younger children, individualized means may be necessary in the future, such as diffusion optic technology (DOT) lenses [52], dual-focus soft contact lenses [53], enhancing the Jesson factor, a small BOZD design, and increasing the frequency of 0.01 % atropine or the concentration of

Table 3
Comparison of IOP within and between groups at different time points.

IOP	DIMS + AT	OK + AT	SV + AT	p1
Baseline	17.38 ± 1.84	17.14 ± 2.33	16.58 ± 2.12	0.061
6-month	17.46 ± 1.72	17.15 ± 2.18	16.43 ± 1.94	0.058
12-month	17.26 ± 1.97	17.20 ± 2.35	16.92 ± 1.86	0.283
6-month-Baseline	−0.05 ± 1.81	0.01 ± 1.74	0.00 ± 2.29	0.978
12 month-Baseline	−0.02 ± 2.64	0.06 ± 2.17	0.20 ± 1.71	0.563
p2	0.81	0.96	1.00	
p3	0.96	0.89	0.31	

p1: ANOVA was used for comparisons between groups.
p2: Paired *t* tests were used for comparisons within groups (6 months vs. baseline).
p3: Paired *t* tests were used for within-group comparisons (12-month vs. baseline).
DIMS + AT, 0.01 % atropine combined with defocus-incorporated multi-segment spectacles;
OK + AT, 0.01 % atropine combined with orthokeratology;
SV + AT, 0.01 % atropine combined with single-vision spectacles;
IOP, intraocular pressure.

Table 4
Comparison of AMP within and between groups at different time points.

AMP	DIMS + AT	OK + AT	SV + AT	p1
Baseline	14.72 ± 1.79	14.64 ± 1.65	14.61 ± 1.07	0.23
6-month	12.71 ± 1.27	12.48 ± 1.68	12.73 ± 1.15	0.10
12-month	12.84 ± 1.28	12.76 ± 1.26	12.79 ± 0.99	0.34
6-month-Baseline	−1.93 ± 1.50	−1.84 ± 0.80	−1.82 ± 0.48	0.09
12-month-Baseline	−1.88 ± 1.77	−1.88 ± 0.72	−1.80 ± 0.81	0.11
p2	<0.001	<0.001	<0.001	
p3	<0.001	<0.001	<0.001	

p1: ANOVA was used for between-group comparisons.
p2: Paired *t* tests were used for comparisons within groups (6 months vs. baseline).
p3: Paired *t* tests were used for within-group comparisons (12-month vs. baseline).
DIMS + AT, 0.01 % atropine combined with defocus-incorporated multi-segment spectacles;
OK + AT, 0.01 % atropine combined with orthokeratology;
SV + AT, 0.01 % atropine combined with single-vision spectacles;
AMP: amplitude of accommodation.

Table 5
Comparison of corneal staining between groups.

Corneal pitting	DIMS + AT	OK + AT	SV + AT	*P*
None	160 (98.81 %)	88 (84.62 %)	73(97.34 %)	<0.001
Grade I	2 (1.19 %)	13 (12.5 %)	2 (2.66 %)	<0.001
Grade II	0.00(%)	3 (2.88 %)	0.00 (%)	0.032

DIMS + AT, 0.01 % atropine combined with defocus-incorporated multi-segment spectacles.
OK + AT, 0.01 % atropine combined with orthokeratology.
SV + AT, 0.01 % atropine combined with single-vision spectacles.

low-concentration atropine drops, as well as posterior scleral reinforcement.

4.1. Limitations

However, this study has several limitations. First, this study was retrospective, and therefore, the three groups were not identical at baseline for a single treatment at the time of enrollment; thus, a linear mixed model was used to adjust for age, sex, SE, and AL at baseline. Second, the SER change could not be determined by wearing an Ortho-K; therefore, only AL was observed in the OK + AT group. Third, the Ortho-K in the study included VST and CRT designs, but individualized designs such as small BOZD or increasing Jessen factors were not included. Finally, this study only collected data for one year, and the long-term effectiveness and safety need to be further explored.

5. Conclusion

Controlling low myopia in younger children is a significant challenge. In this 12-month retrospective study, the combination of DIMS or Ortho-K together with 0.01 % atropine produced similar reductions in myopia progression in children with SER between −0.5 and −3.0 DS aged between 6 and 11 years. More RCTs are needed to confirm these results.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Authors' contributions

Lei Guo and Cheng Peng conceived and supervised the study;

Lei Guo, Cheng Peng, Xindan Cao and Ziqi Guo designed the experiments;

Lei Guo, Zhiyuan Wei, Hongfei Ming, Yue Zhao, Yue Zhang, and Ziqi Guo collected the data and performed the experiments;

Lei Guo, Cheng Peng and Bing Ma analyzed the data;

Lei Guo, Cheng Peng, Xindan Cao, Ziqi Guo and Zhiyuan Wei wrote the manuscript;

Lei Guo and Cheng Peng revised the manuscript.

All the authors reviewed the results and approved the final version of the manuscript.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.clae.2024.102317.

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